SAFE TEST PLAN

Testing the D1 on the Cerbo, one safe step at a time

Each stage only adds a little risk, and only after the stage before it has passed. Stages 1–4 can't change what the D1 does at all, because the Cerbo sends it nothing. **Stop at the first thing that looks wrong.**

What protects you

1. **The D1 itself:** it stops on 12 V side over 16–18 V or under 8 V, 48 V side over 58–60 V or under 24 V, over 213 A, a short, or a board temperature over 120 °C (D1 manual).
2. **Your BMS and fuses** on HV+, LV+ and KL30.
3. **The driver's limits** in config.ini, and the **test timer**.
4. **KL15: your emergency stop.** KL15 off shuts the D1's controller down immediately, whatever the Cerbo is doing.

Before you start

- [ ] Fuses on HV+ and LV+ (D1 manual), and 3 A on KL30
- [ ] **KL15 through a switch you can reach** (or a fuse you can pull)
- [ ] A multimeter, plus a DC clamp meter if you have one
- [ ] Know your batteries' maximum charge voltages (12 V chassis and 48 V)
- [ ] The Cerbo's battery monitor in view (48 V current)

Stage	What	Risk
1	Simulation on the Cerbo	None: no converter, nothing sent on CAN
2	Wiring check, everything off	None: no power
3	Listen only	None: the Cerbo sends nothing
4	Installed in Automatic	None: the Cerbo sends nothing, and the D1 works as before
5	First commands with guard rails	Low: small limits, 5-minute timer, KL15 switch in hand
6	Normal use	Only after stage 5 passes

1 Simulation: no converter at all

In **Cerbo Setup**, choose **9**. The real driver runs on your Cerbo against a **pretend D1**, and **no CAN socket is ever opened**. A real installation keeps running beside it.

- [ ] "**DC-DC D1 (SIMULATED)**" appears in the Cerbo's switch pane, and in VRM's Switches
- [ ] Its web page opens at `http://<Cerbo IP>:8089`
- [ ] In **Automatic** the pretend 12 V side slowly drops. Below 12.5 V, Status changes to **Autonomous** and it charges back to about 13.4 V
- [ ] **Buck**: Status changes to Buck and the 12 V side climbs to your Buck voltage (13.5 V). **Off**: the current goes to zero
- [ ] **Boost**: the 48 V side rises a little, the 12 V side drops, and boost stops at 12.3 V
- [ ] Change the set-points and limits: values outside config.ini are refused

Stop with Ctrl+C. The simulated device disappears. The pretend D1 follows the manual's logic, but it can't tell you how the real one reacts to commands. That's what stage 5 is for.

2 Wiring check, everything off

- [ ] Continuity: RJ45 pin 7 to white/brown, pin 8 to brown, pin 3 to white/green (guide step 2b)
- [ ] **No continuity** (no beep) between any of the three network-cable wires and **LV+** or **HV+**
- [ ] CAN ground (white/green) to the D1 GND terminal: **beeps**
- [ ] RJ45 **not** in the Cerbo: white/brown to brown reads **about 120 Ω** (your resistor at the D1)
- [ ] RJ45 in the Cerbo, with the VE.Can terminator in the other socket: **about 60 Ω**

3 Listen only

Power the D1 as normal (KL30 and KL15 on). In Cerbo Setup choose **2** (find), then **3** (watch). The Cerbo sends nothing.

- [ ] Converter found, at the right speed (if it reports bus errors, switch speed as the guide says)
- [ ] 12 V side reading within about 0.2 V of your meter on the D1's LV+ to GND
- [ ] 48 V side reading within about 0.5 V of your meter on HV+ to GND
- [ ] No alarms, or only alarms that make sense (for example low voltage when the battery really is low)

4 Installed in Automatic

Choose **4** (install). The driver starts in **Automatic** and still sends nothing. Leave it running for a day.

- [ ] Cards on the Cerbo, in VRM and on the web page, with sensible readings
- [ ] The D1 behaves exactly as before: buck when the 12 V side is low, standby otherwise

5 First commands, with guard rails

Prepare: in Cerbo Setup choose **1**, set these test values, save, then choose **4**:

config.ini line	Test value	Why
`test_timer`	**5**	Buck/Boost switch to Off after 5 minutes
`buck_current_limit_max`	**20**	Small current while testing
`boost_current_limit_max`	**10**	Small current while testing
`boost_voltage`	Your 48 V battery voltage **now + 0.5 V**	Boost can only nudge it
`boost_voltage_max`	Your 48 V battery voltage **now + 1 V**	Nobody can set more

Conditions: 48 V side between 41 and 58 V; 12 V batteries healthy; someone at the **KL15 switch**; a meter on the D1's 12 V terminals. Do each step, watch for 1–2 minutes, and fill in the table.

#	Choose	Expect	What you saw	OK
5a	**Off**	Status: Standby/Idle. 12 V current about 0		
5b	**Automatic**	Status: back to the D1's own behaviour within a few seconds		
5c	**Buck** (13.5 V, ≤ 20 A)	Status: Buck. 12 V current ≤ 20 A (check with the clamp meter, and note which way is +). The 12 V side rises toward 13.5 V and never above about 13.7 V		
5d	**Automatic**	Back to the D1's own behaviour		
5e	**Boost** (now + 0.5 V, ≤ 10 A)	Status: Boost. 48 V current ≤ 10 A. The 48 V side rises a little and stops at the target. The 12 V side drops (it's the source). After 5 minutes the test timer switches to Off		
5f	**Automatic**	Back to the D1's own behaviour		
5g	**Buck**, then unplug the CAN cable	The hand-back test (guide Step 6): does the D1 keep going, stop, or return to its own logic? Plug it back in, then choose Automatic		

> **Stop at once (KL15 off) if:** any current goes above the limit you set; any voltage goes above your batteries' limits; a red **Fault** appears; the board temperature climbs fast (above about 80 °C); or anything smells, gets hot or sounds wrong. Then switch the Cerbo to **Automatic** before you switch KL15 back on.

6 Normal use

- [ ] Set `test_timer = 0`, and set the limits to what your wiring, fuses and batteries can take (option 1, then 4)
- [ ] Put `boost_voltage` back to your 48 V battery's charge voltage
- [ ] Use **Automatic** for everyday running, at least until you know how the D1 handles losing CAN (step 5g)
- [ ] Send Nomadic what you saw in 5g, and which way the current sign went in 5c

Tested and signed off by: _____ Date: _____